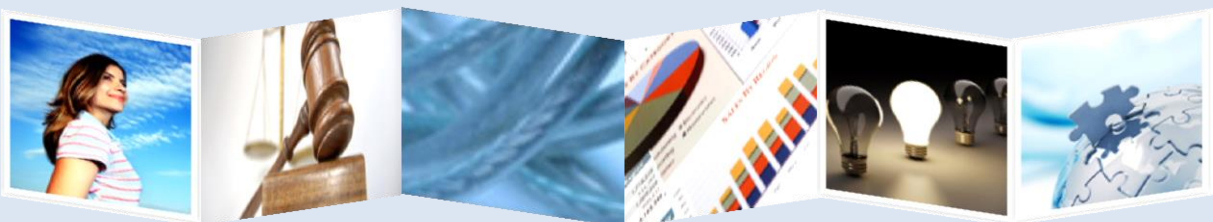
Pre-feasibility study for the establishment of thyme cultivation, production and preparation project


2017



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1. Executive summary

The project aims to cultivate, produce green thyme and to, dry, mix, and prepare it to be ready for domestic consumption. The proposed project will mainly target shopping malls, supermarkets, civil and military institutions in various parts of the Kingdom, with the existence of the chance to expand and export to Arabic markets and to meet part of the growing demand of Arab Minorities in European markets.

The project will provide 15 new jobs, 6 indirect employees, 9 direct employees, and 6 seasonal employees, where the project will raise the living standard of some families in the governorates by providing a monthly income for workers. The estimated investment cost for the project is about 111 thousand Jordanian dinars, which was determined by the fixed assets that include; the cost of primary agriculture, irrigation systems, machinery, equipment, furniture, transportation, expenses before running, and working capital. Where (40%) of the project will be funded through loans and the rest through self-financing.

Financial analysis results indicate that the net value of the project's cash flow is positive, which is an indicator of the financial feasibility of the project, moreover the internal rate of return was higher than the weighted average of the capital cost, which is an indication of the feasibility of the project, also other financial indicators show that the project is profitable for the investor.

Table1 Study major results

Indicator	
Total Investment Requirement (JD)	111,009
Internal Rate of Return (IRR)	%29.17
Pay-Back Period (year)	4
Net present value (JD)	289,418
Job Opportunities provided by the project	15
Annual revenue projection	1,959,323
Total Operating Expenses (10 years)	1,172,922
Net Profit (10 years)	350,972

2. Key Highlight of Jordan

Jordan Hashemite Kingdom area is 89342 Kilometers with a population exceed 9.8 Millions distributed among 12 Governorates.

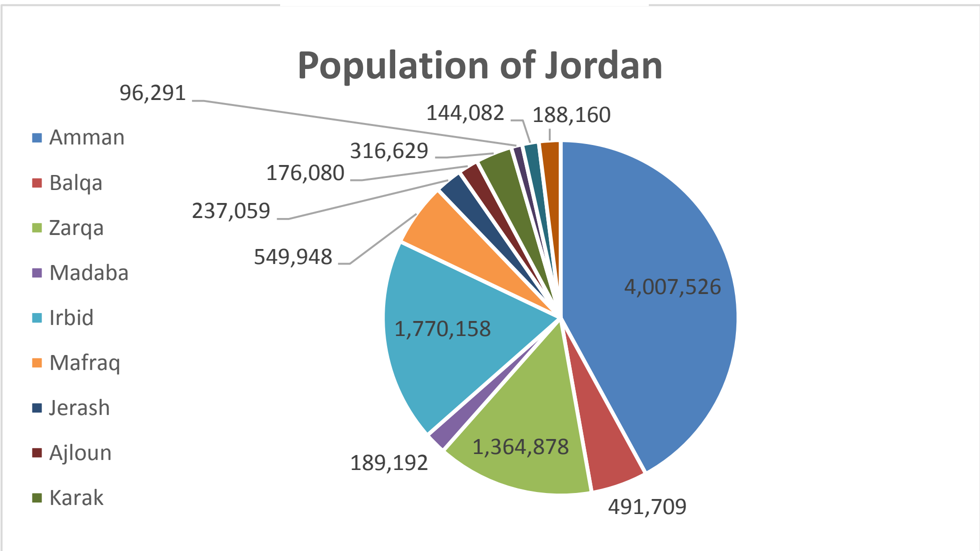
Jordan is bounded by Saudi Arabia, Iraq, Syria and Palestine from the South, East, North and west respectively. Jordan climate is mostly arid desert with rainy season between November to April Months.

Jordan is a free market economy, ranked as the fifth freest economy in the MENA region in the Index of Economic Freedom. Its free market economy enjoys strong partnerships amongst its neighboring country as well as Europe and the USA. Jordan is a signatory of several bilateral and multilateral trade agreements such as (Free Trade Agreement with the US and the EU, a Free Trade Agreement with the EFTA states, an FTA with Singapore, A member of the Greater Arab Free Trade Agreement (GAFTA) and a signatory of the Aghadir Agreement. Looking at the range of countries these agreements cover and facilitate trade between, one is confident to say that Jordan has business gateways and partnerships across the globe.

Population Overview

Based on the latest surveys done by Department of Statistics (DOS) in 2017, total number of population has reached around 9.814.995 with an increase rate of 2.88% from 2016 year, figure below summarizes the population distribution among the kingdom governorates.

Population of Jordan :Figure 1



Department of Statistics (DOS), 2015

Moreover, and based on the latest survey conducted by (DOS) which related to population nationality distribution over the governorates, it was found that total Non-Jordanian residents constitutes around 30% of the total population. Half of these are Syrians (1.3 Million) concentrated mainly in Amman Capital (436 Thousand) then Irbid (343 Thousand), Mafraq at 208 Thousand and Zarqa at 175 Thousand.

Jordan prides itself in its youthful population with 35% of ages below 15 years old and only 4% with ages above 65 years old. Table below summarizes population distribution of Jordanians according to Age Group:

Table2 Population Distribution according to Age Group:

Age Group	Population %	Males	Females
0-14 Years	%35.04	%19.2	%18.2
15-64 Years	%61.02	%30.7	%28.6
More than 65 Years	%3.94	%1.6	%1.6

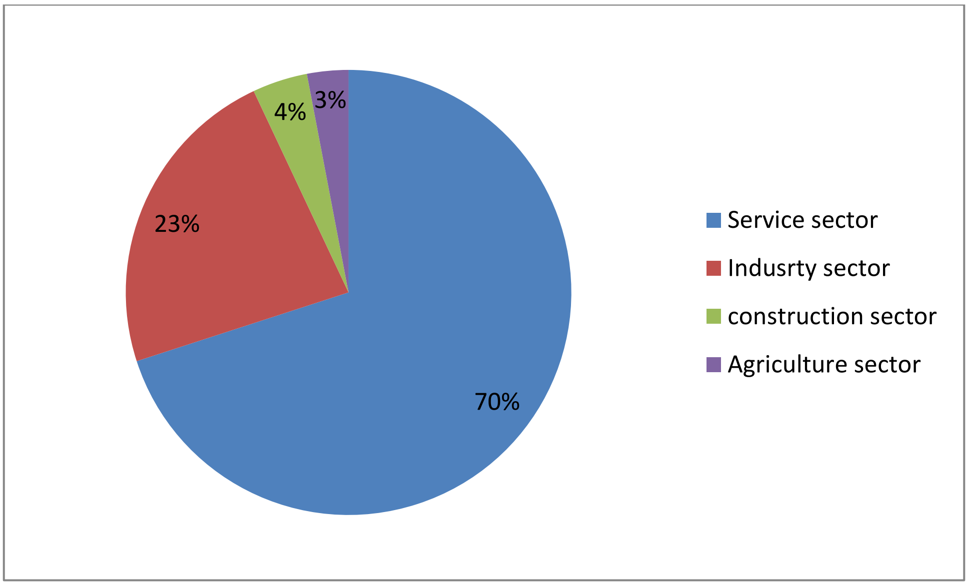
Department of Statistics (DOS), 2015.

Overiew of Jordan Economy

Classified as an upper middle economy by the World Bank and as a country of High Human Development by the UNDP, Jordan is an important financial power in the Middle East. Its small market witnessed growth in the last two decades since King Abdullah II accessed the throne. Jordan's GNI per capita has increased in the last four years due to the basket of economic reforms that were enacted in the recent decade such as (the new Income Tax Law, Public Private Partnership Law and Investment Law). These laws aim to encourage the participation of the private sector in the Kingdom's economic development and provide an enabling legislative environment for current and new investment opportunities.

The Jordanian economy is overwhelmingly services oriented and its contribution in the GDP is 68.1%. The contribution of the industrial sector is 22.4%, the construction sector is 4.2% and the agricultural sector is 2.9%. These contributions are aimed to increase to (27.4%), (5.8%) and (3.4%) respectively and that of the service sector is due to decrease according to Jordan 2025 vision.

Figure 2: Sectors Contibution to Jordan's Economy



Jordan's exports include variety of textiles, potassium, phosphates, fertilizers, vegetables and pharmaceutical products. While it's main imports are crude and refined petroleum. Distinguished by its strategic location, on the crossroads between Asia, Africa and Europe with strong connections to the Levant and the GCC, Jordan has a regional market of interest that represents US$3.8 trillion market and compromising 380 million consumers.

Jordan infrastructure ranks comparatively well (38th out of 148 comparable economies), with an extensive 8000 KM road network connecting Jordan domestically and externally The new Queen Alia International Airport and the Port of Aqaba are the major gateways to the international market. In addition to some mega projects such as the Red- Dead Sea Canal and the national railway network that will be developed to position Jordan as a hub for regional commerce.

Jordan banking system is quite sophisticated, resilient and in compliance with international standards, making it very attractive and trustworthy to investors. This is reflected in the fact that 50% of equity in licensed banks in Jordan is held by non-Jordanians, and non residents' deposits in Jordanian banks witnessed a steady growth of 19.2%.

Moreover, realizing the value of MSMEs to drive economic growth, the government has developed the national Strategy for the encouragement of entrepreneurship and the development of micro small and medium sized enterprises for 2015-2019. This shows the commitment of the Jordan government to enhance the private sector development and leveraging the country's strong human capital.

Jordan prides itself in its youthful population. It's the country most valuable capital. A tech savvy well educated and trained workforce attracts a lot of investors to Jordan. More than 20.4% of Jordan's GDP is dedicated to the education and capacity building for the labor force, this has resulted in securing a 91% literacy rate and enabled Jordanians to be among most hired and qualified middle-eastern workforce.

Such solid, diverse and resilient characteristics of the Jordanian economy and investment scene position Jordan as a competitive investment destination.

Major Economic Indicators

- **GDP Growth**

The Jordanian economy slowed and reached 2.4 percent in 2015 compared to growth rate of 3.1 in 2014 and similar to MENA growth rates.

The growth also slowed for trade, restaurants and hotels; manufacturing; transport; and electricity and water. Meanwhile, finance, insurance, and business services advanced at a faster pace.

- **Inflation rate**

Inflation rate decreased to 0.9 in 2015 after 2% decline in the previous year. Inflation Rate in Jordan averaged 3.1 percent from 2011 until 2015.

- **Unemployment**

The unemployment rate in Jordan was recorded at 13 percent in 2015, comparing to 11.9 percent a year earlier. This increase due to the current instability in the labor market structure as well as high competition from low cost foreign labors especially from Syria.

- **External Sector**

A number of risks manifested in 2015, the closure of land trade routes with Syria and Iraq and the deepening instability in the region adversely impacted many external sector indicators such as trade, tourism and direct Investment.

Moreover, loans disbursements increased by JD 545.3 million, due to the use of the International and Arab Monetary Funds (IMF and AMF) credit facilities. As an outcome of these developments, the overall balance of the balance of payments registered a surplus of JD 328.7 million in 2015, compared with a surplus of JD 1,550.7 million in 2014.

Further, net international investment position (IIP) witnessed an increase in the Kingdom's net obligations to abroad; to reach JD 24,357.5 million, compared to a net obligation of JD 22,578.8

million at the end of 2014 as a result of the increase in the stock of external financial assets and liabilities of all resident economic sectors to reach JD 18,657.9 million and JD 43,015.5 million; respectively.

External Debt in Jordan increased to 9390.50 JOD Million in 2015 from 8030.10 JOD Million in 2014. External Debt in Jordan averaged 5433.67 JOD Million from 1988 until 2015, reaching an all time high of 9390.50 JOD Million in 2015 and a record low of 3640.20 JOD Million in 2008.

Investment Climate in Jordan

Investment in Jordan is considered as one of the vital sources to boost the economy considering that Jordan's economy is among the smallest in the Middle East, with insufficient supplies of water, oil, and other natural resources, underlying the government's heavy reliance on foreign assistance.

The country's location, supported by myriad free trade agreements (FTAs) offering access to 1.5bn consumers, enables the kingdom to be a strategic trade route to many of its neighboring countries and regions.

Jordan aspires to create a competitive investment destination capitalizing on its many advantages mentioned above. The government focused its efforts to implement significant advances in structural and legal reform. These efforts are represented in the new Investment Law of 2014, the tax law, in addition to other endeavors related to providing greater access to credit for MSMEs, and by providing greater investment incentives to specific priority sectors identified by the new Investment law and the Jordan 2025 vision.

Furthermore, in it endeavor to enhance the business environment, several geographically industrial states were created across the kingdom. These range in type to include (industrial estates, free zones and special economic zones). Under the new Investment Law, these zones are given a number of incentives that include a tax rate of 0% on exports, sales tax, import duties, social service tax, dividends tax and a 5% income tax from all economic and manufacturing activities undertaken in the development zones. As for the investments in Free Zones, they enjoy Exemptions from customs duties, income tax exemptions and exemptions from land and building taxes among others. Both development and free zones also enjoy facilitations regarding visa and residency permits for investors and workers in addition to Repatriation of capital and profits in a convertible currency. Such estates have succeeded in attracting relatively large amounts of FDI to Jordan.

Key National Invesment Priorities:

Jordan Investment commission worked on taking a leading role in the application of government policies to promote and attract domestic and foreign investments and create an investment environment that stimulates economic performance through investment promotion strategy launched in 2016

Jordan investment commission aims to:

- Regulating the special provisions governing Development Zones and Free Zones in the Kingdom and developing them placing them in service of the national economy as well as monitoring their functioning.
- Developing plan and programs to stimulate domestic and foreign investments.
- Establishing trade centers and organizing exhibitions as well as opening markets and organizing trade missions in order to promote national products, in addition to marketing and development of national exports and encouraging investment.
- Taking appropriate decisions related to private or public institutions to improve Investors' confidence in Jordan's investment environment.

3. Market analysis

Project description

The project idea is to plant 50 donums of thyme using exposed agriculture and drip irrigation, and to establish a workshop to prepare ready flavoured thyme using machinery, equipment and specialized crushers. Such projects are agricultural projects that are directly related to manufacturing and marketing, and could be established easily in case of availability of water, where these kind of projects do not require high capital, moreover they are family nature, where all family members can collaborate in the operation and management of this project.

Project objectives

Jordan market needs thyme a lot, where the imports average value for the years 2012-2015 is approximately 3.1 million dinars. The project aims to:

1. Achieve profitable return to the project owners.
2. To cultivate, produce and supply thyme with high quality.
3. Provide job opportunities in less developed regions

4. Export Jordanian flavored thyme to Arabic and foreign markets with a distinctive brand.

Proposed product

Thyme is considered as a traditional food in our country and people used to collect it from mountain regions in spring and summer, it has many uses as foodstuff, where green thyme is grinned after getting dried and transformed into flavored thyme mixture by adding wheat, Sesame and sumac, spices, olive oil and salt. As well as filling and packaging it in suitable containers 1/2 kg to 1 kg, so the idea of the project is to produce thyme of high quality and export it to Arabic and European markets through foreign agreements.

Targeted segments

- Jordan families (local market)
- Export (external market)

Market size analysis

The data of the Department of agriculture indicate that the imported quantities of green thyme (non-dried) to central markets (114) tons and the area planted with thyme is (2175) donum, (1700) domun are planted in high land areas and (445) donum are planted in the Jordan Valley area during 2015. The productions of thyme of these planted areas are inadequate and don't meet population's needs, allowing traders to import thyme from the West Bank, Lebanon and Syria to cover market needs. Moreover, the number of farmers who work in this sector is limited, therefor; many non-governmental organizations are promoting or supporting families financially to produce thyme by intensive or semi-intensive methods.

The following table shows the value and quantity of imports of thyme during the time period (2012-2015) , where the average amount of imports of thyme alone (1022) tons with a value of about (3.1) million JD, the table also shows that there is a deficit in the balance of trade of thyme around the year 2014 where it is equal to(416) thousand JD.

Table 3: Value and quantity of imports, exports, re-exports and trade deficit of thyme

Year	Value of (JD) imports	Quantity of (Ton) imports	Value of exports (JD)	Quantity of exports (Ton)	Value of re-exports: (JD)	Quantity of re-exports (Ton)	The total value of exports (000)	Value of trade deficits (JD)
2012	1,734	651	2,460	2,126	28	38	2,488	754
2013	2,448	931	3,198	2,664	57	28	3,255	807
2014	4,144	1,233	3,665	2,842	63	65	3,728	-416
2015	4,230	1,273	4,482	3,241	171	139	4,653	424
Average	3,139	1,022	3,451	2,718	80	67	3,531	392

Source: Department of statistics (2017) external trade bulletin.

Domestic demand of thyme mixture was estimated based on the household expenses study for 2013 published by Department of statistics, where the estimated consumption per capita was about (220) g, annually. This average varies by governorates (previous table). And by looking at the census Bulletin of2015 assuming the annual population growth rate is 2.2% for the next period. As indicated by the Department of statistics; a population projection was made during the period (2018-2027) and then multiplied by the expected population with an average per capita consumption of thyme (assuming steady depreciation rate), future demands for thyme was estimated during the period (2018 – 2026) as shown in the previous table.

Table 4: Per capita consumption of thyme, expected population and estimated demand to 2017

Administrative Distribution	population Inhabitants/201	Average per capita (Kg/year) consumption	Expected (Ton) demand
Amman governorate	4,197,886	0.25	1,049
Balqa' governorate	515,035	0.1	52
Zarqa'a governorate	1,429,690	0.22	315
Madaba governorate	198,139	1.08	214
Irbid governorate	1,854,168	0.16	297
Mafraq governorate	576,033	0.12	69
Jerash governorate	248,274	0.07	17
Ajloun governorate	184,456	0.1	18
Karak governorate	331,624	0.28	93
Tafelieh governorate	100,897	0.2	20
Maa'n governorate	150,928	0.12	18
Aqaba governoarete	197,094	0.32	63
Jordan	**9,984,223**	**0.22**	**2,225**

Assuming that there will be no increase on the average rate of demand on thyme, and the increase will be only within the limits of the annual increase in the population growth rate of 2.2%, and there is an additional demand on thyme from bakeries and restaurants which was estimated at approximately 500 tons annually as a result of the increased number of bakeries and restaurants that offer thyme pastries. (where it is difficult to determine the percentage of restaurants and bakeries demand due to the lack of precise specialized data); it is expected that the future demand for the years 2017-2026 as shown in the previous table during the next period (2017-2027), 2725 tons of 2017 to about 3,302 tons in 2026.

Table 5: Estimate expected demand for thyme (2018 – 2027) in tons

Administrative Distribution	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Amman governorate	1,049	1,073	1,096	1,120	1,145	1,170	1,196	1,222	1,249	1,277
Balqa' governorate	52	53	54	55	56	57	59	60	61	63
Al-Zarqa'a governorate	315	321	329	336	343	351	358	366	374	383
Madaba governorate	214	219	224	228	233	239	244	249	255	260
Irbid governorate	297	303	310	317	324	331	338	345	353	361
Al-Mafraq governorate	69	71	72	74	75	77	79	80	82	84
Jerash governorate	17	18	18	19	19	19	20	20	21	21
Ajloun governorate	18	19	19	20	20	21	21	21	22	22
Al-Karak governorate	93	95	97	99	101	104	106	108	111	113
Al-Tafelieh governorate	20	21	21	22	22	22	23	23	24	25
Maa'n governorate	18	19	19	19	20	20	21	21	22	22
Al-Aqaba governoaret	63	64	66	67	69	70	72	73	75	77
Total domestic demand	2,225	2,274	2,324	2,375	2,428	2,481	2,536	2,591	2,649	2,707
Restaurants and bakeries	500	500	511	522	534	545	557	570	582	595
Total demand	2,725	2,774	2,835	2,897	2,962	3,026	3,093	3,161	3,231	3,302

Competitors analysis

The estimated domestic production of thyme is about 2635 tons, and it is cultivated in many green houses that produce thyme under protected cultivation system, they are estimated to be about 363 houses with an area around 181 donum, while the rest is produced by uncovered agriculture system. The local market needs are met by about 1362 tons of locally produced thyme which is estimated based on the amount of planted acreages of (545) donum in 2015, considering that the production rate is 2.5 ton per donum / year in addition it non-registered local production (wild and domestic) is estimated at approximately 250 tons, in addition to the annual imported quantities amounted to (1022) ton for the years 2012-2015. Thus the annual demand on thyme equals to (2635) ton.

In the past, thyme was imported from Syria, Lebanon, and Turkey, recently dry thyme is imported from the West Bank with a quantity equals to (680) tons/year. Ready thyme is then exported to Saudi Arabia, Qatar, Bahrain, United Arab Emirates and Denmark, US as well as some European countries with limited quantities ranging between 2-5 tons/year. Therefore, the proposed project will be dedicated to promote the export and to meet the future demand in the domestic market. Given that the recent trend shows an increase in the amount of imports of this substance.

There are a number of producers in the market such as; Al-herbawe, Al-shahbaa, Al-safa, Al-skafe group and Majatek company. Where some cooperative societies from the West bank have newly entered the market offering their products in the malls. So the project is not expected to face difficulties in achieving expected sales as thyme is considered as locally desired Food commodities and at the local and Arabic level due to its nutritional significance. Jordan still needs thyme either for local consumption or for export. Thyme is also imported to cover part of the domestic market needs, where the volume of exports should be expanded by opening new export markets with the need to adhere to the quality of the products. Products can be marketed to the domestic market or export to European and Arab countries.

Price analysis and pricing policy

Thyme is sold at malls, perfumery shops and retailers, selling price ranges between (3.5) JD/kg for bad qualities, to (5) JD/kg for specialized thyme, and (6-7) JD/kg for majestic thyme which is the best quality. The differences in the quality of the product mainly, portion of dry thyme in the mixture, and other added spices such as; sumac, Sesame, and toasted wheat.

The project is looking for the production of high quality thyme which complies to the Jordanian specifications where the proportion of thyme should not be less than 35% of total processed thyme and the use of high quality of production inputs , while maintaining good health conditions of manufacturing and storage, so it is suggested that the sale price of one kg of high quality thyme is not less than (3.5) JD per one kilograms for wholesale price for dealers, perfumery shops and retailers and for (5-6) JD/kg price sale for the consumer.

Market share and production capacity

The Project market share is estimated at 50 ton/year from planting 70 donum of fields based on the projected productivity that is estimated about (2.5) ton/year of green thyme , the ratio of dry material to green material(1:10) as reported by professionals in this area. And the market share was estimated according to the following assumptions:

1. Rate of agricultural productivity is 2.5 tons/donum annually.
2. Dried thyme to green thyme ratio equals to 10% of the total green product weight.
3. Ratio of dry thyme to flavored thyme after preparation 35%
4. The productive life of the thyme plant is five years and then new seedlings are planted.

5. Thyme Seedlings are planted at a distance of 0.5 m between one seedling and the other, and 0.7 m between one line and the other, where the number of cultivated plants in the greenhouse reach1250 seedlings/per house and 2500 seedlings/donum.

Project market share can be estimated based on the project's expected production capacity amounting to 1.85% of the market share in the first year, decreases to 1.6% in subsequent years due to the increase of the population and the steady volume of the project.

Expected revenues analysis

The Thyme grown within a plastic houses is harvested at a rate of 5-6 times a year, the planted thyme productivity rate is about 2.5 tons/donum. Thyme is considered perennials but it is preferably to be replanted every five years where productivity starts to drop down. The Wholesale prices range between 3.5-6 JD/kg, the following table shows the expected revenues of the project.

For the purposes of this study, it was considered that the average selling price is JD 3.75/kg after processing, grinding, mixing and packaging, where prices are expected to increase by 2% annually, based on the wholesale prices and consumer prices after deducting profit margin for retailers and wholesalers, where it is expected that the sale price of this product ranges between (5.5-6) JD/ kg. In general, when estimating the sale price for financial analysis it was taken into account that the price should be very competitive, where the product will compete with the existing producers in the domestic market. As shown in the table below:

Table 6: Projected annual revenue for the years (2018-2027)

Project revenues	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Total sales (ton)	**25**	**50**	**50**	**50**	**50**	**50**	**50**	**50**	**50**	**50**
(JD/ton) Product price	3,750	3,825	3,902	3,980	4,059	4,140	4,223	4,308	4,394	4,482
Total sales (JD)	93,750	191,250	195,075	198,977	202,956	207,015	211,155	215,379	219,686	224,080
Total revenues (JD)	93,750	191,250	195,075	198,977	202,956	207,015	211,155	215,379	219,686	224,080

4. Technical analysis

Thyme is considered a perennial plant which is collected in the case of protected agriculture 5-6 times a year and twice a year in the case of exposed agriculture.

Agriculture:

Thyme breeds by transplantation or by seeds from previous season, where its storage period shouldn't exceed 3 years to maintain its vitality, whereas, seeds are planted in spring and summer and are transferred to the permanent fields when seedlings height reach 10-15 cm. transplants are taken from mature plants with 10 cm length, treated with rooting hormone, planted in a suitable environment and then transferred to a permanent place after 45-60 days from transplanting.

Agriculture date:

Thyme seedlings are planted in permanent ground from early spring till the end of autumn while cultivation is avoided in frost days.

Agriculture method:

The land is prepared for planting by removing grasses, stones and the remains of the previous crop and then plowed twice with different vertical angles.

A Fermented organic fertilizer is added by rate of four cubic meters per donum, where non-soluble compound fertilizers are added byrate of 30 kg/donum, mixed well within the soil and then the land is settled. Afterwards the land should be divided into lines where the distance between each line and the other will be 80-100 cm, land is irrigated and after it is half-dried, seedlings are planted at a depth of 10 cm so that the distance between seedling and the other 50 cm, seedlings are watered immediately after planting.

Irrigation:

In the first period of cultivation thyme needs sufficient amounts of water avoiding drought exposure, where it depends on weather conditions, soil quality, and relative humidity. In general, thyme can be irrigated once every 4 days in non-rainy winters. The soil need for irrigation can be determined through testing soil moisture at 5 cm depth around the plant and then compressed by hand, if it sticks together then irrigation is not needed, otherwise, it is needed.

Table 7 expected market share in the local market during (2018-2027)

Year	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Expected population (000)	9,984	10,204	10,428	10,658	10,892	11,132	11,377	11,627	11,883	12,144
Annual per capita consumption (kg/capita)	0.220	0.220	0.220	0.220	0.220	0.220	0.220	0.220	0.220	0.220
Domestic consumption of the product (tons)	2,197	2,245	2,294	2,345	2,396	2,449	2,503	2,558	2,614	2,672
Request for restaurants and bakeries (tons)	500	501	502	503	504	505	506	507	508	509
Domestic demand for virtual goods (tons)	2,697	2,746	2,796	2,848	2,900	2,954	3,009	3,065	3,122	3,181
The expected output of the project (tons)	50	50	50	50	50	50	50	50	50	50
The market share of domestic demand (%)	1.85	1.82	1.79	1.76	1.72	1.69	1.66	1.63	1.60	1.57

Table 8: Sale prices and expected income during the period (2018 – 2027)

Project revenues	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Total sales (tons)	25	50	50	50	50	50	50	50	50	50
The price of the product (JD/ton)	3,750	3,825	3,902	3,980	4,059	4,140	4,223	4,308	4,394	4,482
Total sales (JD)	93,750	191,250	195,075	198,977	202,956	207,015	211,155	215,379	219,686	224,080
Total project revenue (JD)	93,750	191,250	195,075	198,977	202,956	207,015	211,155	215,379	219,686	224,080

Fertilization:

Thyme is considered one of the plants that weaken the soil, where in addition to fertilizers that are added when preparing the soil for planting; compost (dung) fermentation should be added yearly after the last harvest at the beginning of winter, in a rate of 4 cubic meters per donum.

Chemical fertilizers are added as follows:

25 kg ammonium sulfate or 10-15 kg urea/donum after each harvest

20-30 kg superphosphate/donum, added two times, one is added at the beginning of spring and the second one two months later.

20-25 kg potassium nitrate/donum, added once in spring.

Fertilizer containing nutritional elements (iron, zinc, copper) can be sprinkled at a rate of 500 g/m3) after two months of agriculture, and should be repeated every 6 months.

Weeds resistance:

Artificial fertilization shouldn't be added unless getting rid of exotic herbs manually where these herbs compete the nutrients and light and weakens growth.

Harvest:

During the first four months of cultivation thyme should be cut at a height of 7 cm to obtain the largest possible number of buds. Harvest of thyme, can be done 3 times a year in the protected agriculture depending on weather conditions, fertilization and irrigation.

Drying:

Drying is considered as an important process to maintain the active substance in thyme, protect it from corruption and prepare it for storage. Thyme is washed and kept in exposed and shaded placed in thin layers to prevent rot and it should be flipped continually until it dries.

Production capacity:

Each single donum produces 300-500 kg in each harvest , note that the productive life of thyme is about 5 years ,after that it is preferred to renew its cultivation every.

Diseases and pests:

Thyme can be exposed sometimes to certain diseases affecting the amount of the crop and reduce the production of essential oil that gives the taste of thyme, the most important diseases:

Green fly: weeds around plants encourages the growing of the green flies.

Worms: they are Green worms that bites leaves and buds.

Trips: little bug that sucks plant juices from the leaves, causing white spots.

Fungal wilt disease caused by soil fungi, causes plant wilting, can be prevented by soil sterilization before planting.

Required raw materials to produce green thyme

175 tons of thyme green material should be produced and dried to achieve 17.5 ton of dry material. To produce 175 tons of green stuff on the assumption that the single donum productivity is 2.5 tons; it is needed to plant 70 donums of agricultural land using exposed cultivation system and drip irrigation method, where it needs different compound chemical fertilizers, organic fertilizers, pesticides and water needs are, estimated about 600 cubic meters/donum with a purchase price of 0.2 JD/cubic meters. The following table shows the cost of raw material for agriculture and the annual cost to plant 70 donum with thyme.

Table 9 details of the required raw material

Production requirements	Unit	Price/unit	Quantity	Value (JD/donum)	Number of donums	Annual quantity (unit)	Annual cost (JD)
Chemical fertilizers	Kg/donum	0.6	50	30.0	70	3,500	2,100
Organic fertilizers	Ton/donum	50	0.3	15.0	70	21	1,050
Types of pesticides	Liter/donum	50	1	50.0	70	70	3,500
Others	JD/donum	20	1	20.0	70	70	1,400
Water (1)	Cubic meter/donum	0.21	600	126.0	70	42,000	8,820
Total					70		**16,870**

(1) Ground water Extract is sold per hour between farmers which extracts (70) cubic meters of water per hour by (15) JD in the Mafraq Governorate.

Needed raw materials to manufacture thyme

It is expected that the production of the first year would be half of the estimated production of the subsequent years. Which means that only 25 tons of table thyme will be produced in the first year, and will be increased to 50 tons in the second year and subsequent years? Production was estimated based on the ratios of mixing raw materials and its price in the market. The cost of 1 kg of dry thyme was estimated at 4 dinars.

Table 10 Prices quantities and annual cost of raw materials for manufacturing

Production requirements	Price (JD/kg)	Quanti (ty (kg	Cost of 1 kg	Perce ntage	Quantit y in tons	Annual quantity in kg	Annual cost (JD)
Dry thyme in case of purchase (1)	4	350	1.4	%35	17.5	17,500	
Toasted wheat	0.5	250	0.125	%25	12.5	12,500	6,250
Sesame	1.25	250	0.312	%25	12.5	12,500	15,625
Sumac and spices	1.5	130	0.195	%13	6.5	6,500	9,750
Olive oil	3.5	20	0.07	%02	1.0	1,000	3,500
Packing materials	50	1	0.05				2,500
Total (per unit)		1000	2.1	100%	50.0	50,000	37,625

(1)will be produced from the farm and did not enter in the accounts, and used to estimate agricultural area required

Required Human resources

This project needs an agricultural engineer specialized in plant production of medicinal and aromatic plants, in addition to a manufacturing technician, as well as product marketing officer to target markets, agricultural workers, manufacturing workers . As well as the need to temporary employment during the seasons of harvest and transportation it to the workshop. The following table shows the approximate employment and wages requirements:

Table 11 required human resources

Job title	Number	Monthly (JD) salary	Total monthly
Project Manager/agricultural engineer vegetable	1	500	500
Agricultural manufacturing technician	1	300	300
Marketing representative	1	350	350
Guard	1	200	200
Total direct salaries	**4**		**1,350**
Maintenance and repair technician	1	300	300
Agriculture workers	2	300	600
Manufacturing and processing workers	2	350	700
Part time manufacturing labour (1)	2	150	300
Part time labours for the farm (1)	4	150	600
Total of part time labours	**6**		**900**
Total direct salaries	**11**		**2,500**
Total	**15**		**3,850**

Job Description:

- **Agricultural engineer**: Provide the necessary materials for thyme productions, tools and supplies of plant production, the tillage and preparation of agricultural land to be suitable for growing crops such as primary and secondary tillage, applying fertilization to agricultural soil with chemical and organic fertilizer, and processing of pens and installation of irrigation network or irrigation systems in open field. Planting, producing, cultivating, preparing and transporting seedlings, and cultivating thyme seedlings in permanent locations. Irrigating seedlings and fertilizing and calming the seedlings as needed. implementation of measures to prevent and control agricultural pests in accordance with environmental standards. Harvesting of products, assembly and drying of products, processing, sorting and grading of products, packaging in appropriate packaging, filling farm records and business models, documenting expenses, preventive control procedures and schedules, applying occupational safety and health procedures and instructions
- **Food processing technician**: Preparation to receive the raw materials, check the specifications of the raw materials and the appropriate time schedule for their supply, follow up procedures for installation and maintenance of machines and the preparation of appropriate plan for production. The preparation of cleaning programs and disinfection of machines and equipment, Identification of materials allowed to perform cleaning and

disinfection. Preparing the production plan according to the type of food products to be manufactured, quantities of input materials and schedules, determining the number of workers, the machines used, and developing appropriate technical solutions for the problems related to the production process or product quality. Identify the raw materials involved in product composition. In terms of type and proportion of use and follow-up, determine the temperature and humidity suitable for storage of raw materials and food products and the validity for consumption and matching the results of laboratory tests of samples with quality specifications, development of procedures and methods of work and production using modern technologies. Development of quality systems and standards. Preparation of technical reports on food processing operations, implementation of procedures and ensuring occupational safety and health requirements.

- **Marketing Officer**: Preparation for the receipt of products and their inspection, matching their quantities, types and specifications with the purchase order, inspecting their validity for human consumption, filling the forms for the receipt and delivery process, following up the process of loading, downloading and receiving, By defining the types of sales, clarifying the advantages and specifications of different types of products, helping customers to choose their needs, answering their inquiries, securing their requests, following up their processing, Product packaging, customer accounting, invoice editing, inventory of warehouse contents, identification of the plant's needs for agricultural production requirements, preparation of inventories, insurance and storage. Organizing records of sales and financial records, filling out business models and applying the procedures and instructions of occupational safety and health.
- **Agricultural worker / packing and packaging**: Works under the supervision of the agricultural engineer and prepares the packing and packaging tools, and receipt of the crop from the field for drying, sorting and classification of the product, packaging and storage until marketing. Packing and packaging of containers, . Placing the packaging on the production line, sorting the product and removing the distorted and infected parts, classification and grading according to the required specifications. Fill the product in the custom packaging and containers and pack it according to instructions. Collection and packaging of containers, and stored in warehouses or warehouses until marketing.
- **Agricultural worker**: work under the supervision of the agricultural engineer to prepare the necessary materials, preparing the land for cultivating by tillage, leveling, preparing the agricultural land, planting the seedlings of plant, taking care of the crop, spraying the plants with pest control materials such as insecticide and pesticides. Control diseases and rodents

in accordance with environmental standards, disposal of infected seedlings / parts, and protection of the crop from the risk of frost. Harvesting of the products, sorting and grading of products, packing, packaging, storing of products. Apply the safety and occupational safety measures and health procedures

- **Maintenance and Repair Technician**: Works independently or under the supervision of the agricultural engineer, maintains irrigation networks, valves and supplies, ensures water pumping and maintenance of agricultural ponds, repairs and maintenance of machines and equipment from roasters, mixers and general maintenance works in the distillation plant and ensuring its readiness to work, Spare parts and training of workers on how to operate and closing them, and maintenance of warehouses and storage rooms for production of finished goods, implementation of procedures and instructions for safety and health.

Site analysis

Thyme is considered one of the plants that bear different environmental conditions, it withstands temperatures as low as frost, and bear the high temperatures in summer, and whenever less amount of water is given to the thyme the higher amount of volatile oil and thyme keeps its strong scent, where it is not preferred in Jordan to plant thyme in the valley because the high temperature makes the crop lousy and undesirable , so it's preferred to be planted in Mafraq Governorate, thyme is grown by transplantation method, where the suitable time for cultivation inside greenhouses is mid-September, it lasts for five years, based on all these data, it is proposed to establish a number of projects for the production of thyme in Mafraq agricultural areas where surface waters (water harvesting projects) or groundwater are available.

Description of production process

After collecting the plant it dried on shaded ground for 4-7 days , preferred not to dry it under direct sunlight. The proportion green of thyme to dry thyme is (1:10) and according to the Jordan standard (840) thyme mixture has been classified as the mixture resulting from blending all dry thyme leaves , sumac, Sesame, wheat with varying ratios depending on the desired item to be produced, and in most cases dry thyme leaves ratio must equal 30-40% of the mixture. Requirements to produce a mixture of thyme is to be of nice and natural smell and flavor, and it should be , free of live and dead insects or their parts, exotic materials proportion not to exceed (2%), and that no more than 10% moisture. In case it is decideded to invest in this project it is advised to refer to the Jordanian standards of specifications.

The preparation process requires expertise, quality of raw material, considering people's tastes and appropriate manufacturing method to save the taste of thyme components of anise, coriander, cumin, sumac, Sesame and salt in addition to thyme leaves, where it is exposed to the processes of sieving, milling and mixing it with the previous materials according to rates selected by the experience the taste of the manufacturer, where the highest percentage is thyme leaves then wheat and other materials. These materials are toasted and mixed before grinding to get mixed well, and give a flavor and then grind and sieved again to isolate light pieces of sticks and impurities down to packaging and distribution phase

1. Thyme is dried and beaten to remove leaves blades and then transported from the drying field to grinding and mixing workshops.
2. Grinding the thyme by spices mill
3. Adding olive oil and then mixed well
4. spices are roasted and then grinded by spices mill
5. Toasted spices and sesame are added to thyme and then tasted to make sure of the required acidity and salinity and then adjusted according to the required quality

Technical requirements analysis

1.1.1 Land and construction works

The total area needed for the project is about (100) donum, (70) donum for agriculture, and the remaining area for services, facilities, trails, workers' housing, manufacturing workshop, and a place to dry thyme by solar energy. It is suggested to rent the land for the project, and it is expected that the annual rental fee of the donum will not exceed 50 dinars per donum, thus the annual rent value of the land will be JD 5000, and the annual increase was assumed by (2%) on the annual rent.

The project needs the following facilities and buildings: workers' housing, a hall for sorting and grinding thyme, storage rooms, special hall dedicated to mixing, preparing and packing, plus a pond for farm irrigation. The cost includes construction required for the project such as; building, design and schemes, in addition to excavations and surface construction, as shown in the table below:

Table 12: Agricultural and construction required for the project

Construction work	Cost (JD/m²)	Area (m2)	Total (JD)
Small Office and its supplements	40	40	1,600
Workers' housing and its supplements	40	60	2,400
sort ing, roasting and grinding room	50	60	3,000
Pool (2x1010x m) and pipes	3	200	600
Storage warehouse	50	80	4,000
Mixing, preparing and packaging room	50	100	5,000
Site and work adjustment	0.05	70,000	3,500
Planting seedlings (2500 seedlings/donum)	0.035	70,000	6,125
Others	-	-	1,160
Total			27,385

Machinery and equipment

The following table shows the required machinery and equipment required for thyme cultivating and manufacturing:

Machinery and equipment	quantity	Unit cost (JD)	Total cost (JD)
Pond plastic cover and plastic pipes	1	200	200
Drip irrigation system	1	7,000	7,000
manure and tanks	1	400	400
Water pumps	1	1,000	1,000
Plastic sheets for drying	1	200	200
Agricultural tools and equipment	1	500	500
Packing tables	1	500	500
Mixing machine	1	3,000	3,000
spices mill	1	3,000	3,000
vacuumed packaging machine	1	500	500
Electronic balance	2	125	250
Numbering machine	1	200	200
toasting machine	1	500	500
others	1	-	1,725
Total			18,975

Furniture and furnishings

The project needs simple furniture for management and permanent employment, the cost is estimated about 1000 JD, such as simple tables, desks and cabinets.

Transportation and vehicles

The project needs a pick-up to transfer thyme to the drying site, as well as a medium transport vehicle for marketing the products to target markets as are shown in table:

Table 13: Required Transportation and vehicles

Transportation and vehicles	Quantity	Unit cost/JD
Pick-up for internal transportation	1	10,000
Medium transport vehicle	1	12,000
Total	2	

Essential services:

Electricity and water services are available in most parts of the Kingdom, and it is supposed to be available in the project site. The project needs were estimated of necessary services to run the mills and water pumps (electricity, drinking water, diesel and fuel for transportation and marketing). The project requires the availability of irrigation water and access roads that leads to the site, where water for irrigation and road network is available in most kingdom areas. The project also needs 2350 JD for irrigation network maintenance, construction and pump as shown in the following table:

Table 14: Support services required for the project

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Other expenses	7,864	9,910	10,149	10,395	10,646	10,904	11,169	11,440	11,718	12,003
Maintenance expenses	0	3,825	3,902	3,980	4,059	4,140	4,223	4,308	4,394	4,482

Legal requirements and procedures

Agricultural activity in itself doesn't need a license but as the project seeks to produce a prepared food commodity for domestic and to export to Arab and foreign markets prior to the investment starting, a number of legal and administrative conditions should be met. These include legal facility registration, licensing and approvals and others, and deliver the required documents and payment of the initial administrative costs associated with investing

1. Legal registration, whether it was an individual or company Foundation (solidarity, limited partnership, Special contribution, limited liability company, etc.)
2. Registration of the trade name
3. Registration in Chamber of industry/trade. All firms working in the field of industrial or commercial activity must register in Chamber of industry/trade
4. Professional licensing- a prerequisite for legal establishments to be able to do business.
5. Register at the public institution for social security- Each facility is committed under the provisions of Jordanian law to register in social security
6. Food and drug Foundation aapproval is required for handling the product after studying its card, use and components, where food and drug Foundation ensure availability of health requirements to ensure validity of the product for human consumption.

5. Financial Analysis

The Financial Part illustrates all the assumptions which were used when we have developed the study including project expenses related to capital expenditures, operating expenses (Fixed and Variable costs) followed by illustrating project revenues.

Assumptions

Calculation Assumptions:

- The weighted average cost of capital used in the study is 11.97% as it was calculated considering risk free and debt Interest rates as well as Market risk premiums to consider alternative opportunity costs for investor.
- Income tax on the net project income is 0 % during the life time of the project.
- Project working capital has been estimated and added to project cash outflows at the beginning of the project and annual increase in working capital has been also estimated and added to the operating expenses then net working capital has been added to total cash inflows at the end of project lifetime.

The time span for the financial study is 10 years starting from 2018, where annual increase rates have been estimated as per the tables below:

Assumptions

Table 15 General Assumptions

General Information	
All Financial Numbers (Currency)	Jordan Dinars
Financial Study Time Span (Years)	10
Expected Project First Operating Year	2018
Last Year in the Financial Study	2027

Table 16 Currency Exchange Rates

Exchange Rates	Jordan Dinars
American Dollar	0.708
Euro	0.760

Table 17 working time assumption

Working time	
No. of weeks	52
No. of days per week	6
No. of days annually	312
No. of working hours	8

Table 18 Annual Growth Rates Assumptions

Annual Growth Rates	Average
Annual Population Growth Rate	2.20%
Annual Sales Price Increase Rate	2.00%
Annual Expenses Increase Rate	2.00%
Annual Increase Rate in Employees Salaries	3.0%

(1) and (2) Source: Inflation Rates, Central Bank of Jordan

(3) Source: Social Security

Table 19 Expenses Assumptions

Expenses Assumptions	
Raw Materials and Packaging from Total Revenues	0%
Utilities Expenses from Total Revenues	0%
Maintenance Expenses from Total Revenues	2%
Depreciation Expenses from Total Revenues	0.0%
Insurance Expenses from Total Assets Value	0.0%
Marketing Expenses from Total Revenues	3%

Table 20 Income Tax Assumptions

Income Tax Assumptions	Value
(1) Average Income Tax in Jordan	5%
(2) Income Tax Deduction	100%
Income Tax after Deduction	0%
(3) Compulsory Reserve Percentage	10%
Other Assumptions	**Value**
(1) Annual Monthly Salaries have been calculated after multiplying monthly salaries by:	16

Table 21 Risk Premium

Risk Premium	
(1) Risk Free Rate of Return	6.50%
(2) Return to Debt Maturity	7.50%
(3) Market Risk Premium	9.93%
Income Tax Rate	0.00%
(4) Beta	1
Growth Rate	4.00%

(1) Source :Damodaran's Country Default Spreads and Risk Premiums Report

(2) Source :Central Bank of Jordan

(3) Source :Damodaran's Country Default Spreads and Risk Premiums Report

(4) Source :Damodoran Beta By Sector, , pages.stern.nyu.edu

Table 22 Weighted Average Cost of Capital

Weighted Average Cost of Capital	
Average Return on Risk Free Investment	%6.50
Return to Debt Maturity	%7.50
Market Risk Premium	%9.93
Income Tax Rate	%0
Bets	1.00
Equity	83,201
Loans Value	83,201
Loans	
Cost of Borrowing before Tax	%7.50
Cost of Borrowing After Tax	%7.50
Loans Value	83,201
Loans Percentage	%50
Equity	
Cost of Equity	%16.4
Equity Value	83,201
Equity Percentage	%50
Gross	
Project Value	166,403
Weighted Average Cost of Capital	**%11.97**

Capital Expenditures

The estimated cost of the project is 1,354,445 JD and it covers land costs, construction works, machinery and equipment, furniture, pre-operating expenses and working capital. The following tables summarize the details of these expenses:

Table 23 Land Capital Expenditure

Land	(Jordan Dinar/Meters Square) Cost	Area (M2)	Cost(JD)
Required Land	25	600	15,000
Total Cost			**15,000**

Table 24 Construction Work Capital Expenditure

Description	Cost (JOD per Squared Meters)	Area (Squared Meters)	/ Gross Cost Jordan Dinar
Small Office and its supplements	200	200	40,000
Workers' housing and its supplements	150	200	30,000
sorting, roasting and grinding room	150	1,500	225,000
Pool (2x1010x m) and pipes	20	200	4,000
Storage warehouse	100	200	20,000
Mixing, preparing and packaging room	10	500	5,000
Site and work adjustment	-	-	31,900
Planting seedlings (2500 seedlings/donum)			
others			
Total cost			**355,900**

Table 25 Machinery and equipment Capital Expenditure

Machinery and equipment	Number	Unit cost (JD)	Total cost (JD)
Pond plastic cover and plastic pipes	1	200	200
Drip irrigation system	1	7000	7,000
manure and tanks	1	400	400
Water pumps	1	1000	1,000

Plastic sheets for drying	1	200	200
Agricultural tools and equipment	1	500	500
Packing tables	1	500	500
Mixing machine	1	3000	3,000
spices mill	1	3000	3,000
vacuumed packaging machine	1	500	500
Electronic balance	2	125	250
Numbering machine	1	200	200
toasting machine	1	500	500
others	1	-	1,725
Total (JD)			**18,975**

Table 26 Furniture Capital Expenditure

Furniture	Number	Unit cost (JD)	Total cost (JD)
Office furniture	1	1,000	1,000
Workers rest room	1	1,000	1,000
Total			**2,000**

Table 27 Transportation and Vehicles Capex

Transportation and Vehicles	No	Unit Cost/JOD	Total Cost/JOD
Pickup for internal transportation	1	10,000	10,000
Medium vehicle	1	15,000	15,000
Total cost	**2**		**25,000**

Table 28 Pre-Operating Expenses

Pre-operating expenses	Estimated cost (JD)
Governmental fees	1,000
Transportation	1,000
Marketing	2,000
Training	1,000
Total cost	**5,000**

The initial working capital reflects the project's amounts needed to cover all the expenses during operation until the project starts generating income, the following table shows the components of the initial working capital:

Table 29 Working Capital

Working Capital	No. of Months	%	Estimated Cost (JD)
Operating expenses	6	%50	43,366
General and administrative expenses	6	%50	3,740
Marketing expenses	6	%50	1,009
Indirect Salaries	6	%50	10,800
Total			**58,915**

Table 30 Expenses Summary

Capital expenditure summary	Cost (JD)	Contingency	Cost (JD)	%
Land	15,000	10%	16,500	%9.9
Construction works	27,385	10%	30,124	%18.0
Machinery &Equipment	18,975	10%	20,873	%12.5
Furniture	2,000	10%	2,200	%1.3
Vehicles	25,000	10%	27,500	%16.4
Pre-Operating Expenses	5,000	10%	5,500	%3.3
Working capital	58,915	10%	64,807	%38.7
Total (JD)	**152,275**		**167,503**	**100%**

Sources of Funding

The financing structure of the project consists of loans and Equity. The investment cost of this project is estimated at (167,503) JD, where 50% of the project will be financed through loans (83,751) JD, and the remaining (50%) through (Equity) with (83,751) JD. The following table summarizes the general structure of the required financing:

Table 31 Sources of Funding

Funding Sources	Amount	Percentage

Equity	83,751	%50
Loans	83,751	%50
Total	**167,503**	**%100**
Use of Fund	**Amount**	**Percentage**
Capital Expenditures	97,196	%58.03
Pre-Operating Expenses	5,500	%3.28
Working Capital	64,807	%38.69
Total	**167,503**	**%100.00**

Operating Expenses

The following table summarizes the expected operating expenses for the proposed project (2018-2027):

Table 32 Operating Expenses

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Raw Materials for agriculture	8,435	16,870	17,207	17,552	17,903	18,261	18,626	18,998	19,378	19,766
Cost of annual raw materials required for	18,813	37,625	38,378	39,145	39,928	40,727	41,541	42,372	43,219	44,084
Maintenance from Revenues	0	3,825	3,902	3,980	4,059	4,140	4,223	4,308	4,394	4,482
Direct Salaries	51,600	41,200	42,436	43,709	45,020	46,371	47,762	49,195	50,671	52,191
Others	7,885	9,952	10,192	10,439	10,691	10,950	11,215	11,487	11,766	12,052
Total	**86,732**	**109,472**	**112,115**	**114,824**	**117,601**	**120,448**	**123,367**	**126,360**	**129,428**	**132,574**

General and Administrative Expenses

The following table summarizes the expected general and administrative expenses for the proposed project (2018-2027):

Table 33 General and Administrative Expenses

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Telecom	500	510	520	531	541	552	563	574	586	598

Hospitality	200	204	208	212	216	221	225	230	234	239
Stationary and Printing	100	102	104	106	108	110	113	115	117	120
Renting the land	5,000	5,100	5,202	5,306	5,412	5,520	5,631	5,743	5,858	5,975
Training	1,000	0	0							
Miscellaneous	680	592	603	616	628	640	653	666	680	693
Total	**7,480**	**6,508**	**6,638**	**6,771**	**6,906**	**7,044**	**7,185**	**7,329**	**7,475**	**7,625**

Marketing Expenses

Table 34 Marketing Expenses

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Marketing Expenses	**2,018**	**2,058**	**2,100**	**2,142**	**2,184**	**2,228**	**2,273**	**2,318**	**2,364**	**2,412**

Human Resources

The total annual salaries and monthly wages has been estimated on 16 months for the purpose of considering annual increments on salaries as well as payments related to social security and health insurance, etc.

The following table summarizes the expected annual salaries and wages, considering annual increment on salaries as per the income statement:

Table 35 expected number of employees

Job title	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Indirect employees										
Project Manager/plant production engineer	1	1	1	1	1	1	1	1	1	1
Agricultural production technician	1	1	1	1	1	1	1	1	1	1
Marketing representative	1	1	1	1	1	1	1	1	1	1
Guard	1	1	1	1	1	1	1	1	1	1
Total number of indirect employees	**4**	**4**	**4**	**4**	**4**	**4**	**4**	**4**	**4**	**4**
direct employees										
Maintenance and repair technician	1	1	1	1	1	1	1	1	1	1
Agriculture workers	2	2	2	2	2	2	2	2	2	2
Production and processing workers	2	2	2	2	2	2	2	2	2	2
Part time manufacturing labour (1)	2	2	2	2	2	2	2	2	2	2
Part time labours for the farm (1)	4	4	4	4	4	4	4	4	4	4
Total number of direct employees	**11**	**11**	**11**	**11**	**11**	**11**	**11**	**11**	**11**	**11**
Grand total	**15**	**15**	**15**	**15**	**15**	**15**	**15**	**15**	**15**	**15**

Table 36 Expected monthly Salaries and Wages (2018-2027)

Job title	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Indirect employees										
Project Manager/plant production engineer	500	515	530	546	563	580	597	615	633	652
Agricultural production technician	300	309	318	328	338	348	358	369	380	391
Marketing representative	350	361	371	382	394	406	418	430	443	457
Guard	200	206	212	219	225	232	239	246	253	261
direct employees										
Maintenance and repair technician	300	309	318	328	338	348	358	369	380	391
Agriculture workers	300	309	318	328	338	348	358	369	380	391
Production and processing workers	350	361	371	382	394	406	418	430	443	457
Part time manufacturing labour (1)	150	155	159	164	169	174	179	184	190	196
Part time labours for the farm (1)	150	155	159	164	169	174	179	184	190	196

Table 37 Expected Annual Salaries and Wages (2018-2027)

Job title	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Indirect employees										
Project Manager/plant production engineer	8,000	8,240	8,487	8,742	9,004	9,274	9,552	9,839	10,134	10,438
Agricultural production technician	4,800	4,944	5,092	5,245	5,402	5,565	5,731	5,903	6,080	6,263
Marketing representative	5,600	5,768	5,941	6,119	6,303	6,492	6,687	6,887	7,094	7,307
Guard	3,200	3,296	3,395	3,497	3,602	3,710	3,821	3,936	4,054	4,175
Total indirect salaries	**21,600**	**22,248**	**22,915**	**23,603**	**24,311**	**25,040**	**25,792**	**26,565**	**27,362**	**28,183**
direct employees										
Maintenance and repair technician	3,600	4,944	5,092	5,245	5,402	5,565	5,731	5,903	6,080	6,263
Agriculture workers	7,200	9,888	10,185	10,490	10,805	11,129	11,463	11,807	12,161	12,526
Production and processing workers	8,400	11,536	11,882	12,239	12,606	12,984	13,373	13,775	14,188	14,613
Part time manufacturing labour (1)	3,600	4,944	5,092	5,245	5,402	5,565	5,731	5,903	6,080	6,263
Part time labours for the farm (1)	7,200	9,888	10,185	10,490	10,805	11,129	11,463	11,807	12,161	12,526
Total indirect salaries	**30,000**	**41,200**	**42,436**	**43,709**	**45,020**	**46,371**	**47,762**	**49,195**	**50,671**	**52,191**
Total salaries	**51,600**	**63,448**	**65,351**	**67,312**	**69,331**	**71,411**	**73,554**	**75,760**	**78,033**	**80,374**

Depreciations

The following tables illustrate cost of Capex and Depreciation rates:

Table 38 Capex and Depreciation Expenses

Capex Cost and Annual Depreciation Rates	Cost (JOD)	Depreciation Rate	Annual Additions Percentage
Land	16,500	%0.0	%0.0
Construction Works	30,124	%5.0	%0.0
Machineries	20,873	%10.0	%2.0
Furniture	2,200	%10.0	%5.0
Vehicles	27,500	%10.0	%0.0
	97,196		

Table 39 Depreciations and Additions on Construction Works

Capex, Annual Additions and depreciaitions	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Total Fixed Assets	97,196	97,723	98,265	98,820	99,391	99,976	100,578	101,195	101,829	102,481
Total Depreciation Expenses	6,563	6,616	6,670	6,726	6,783	6,841	6,902	6,963	7,027	7,092
Total Accumulated Depreciation	6,563	13,180	19,850	26,576	33,359	40,200	47,102	54,065	61,092	68,184
Total Additions	0	527	541	556	570	586	601	618	634	652
Total Net Book Values	90,633	84,544	78,415	72,245	66,032	59,776	53,476	47,130	40,738	34,297

Loan

The table below summarizes all details related to the loan such as annual instalments and payment methods for the remaining amount of the loan for each year of the project

Table 40 Loan Details

Loan Value	83,751
Annual Interst Rate	9.50%
Loans Period	5
Grace Period	0
Loan Starts at year:	2018

Year	Annual Payment	Interest	Capital	Loan Remaining Value
2018				83,751
2019	21,812	7,956	13,856	69,896
2020	21,812	6,640	15,172	54,724
2021	21,812	5,199	16,613	38,111
2022	21,812	3,621	18,191	19,920
2023	21,812	1,892	19,920	0

JORDAN INVESTMENT COMMISSION هيئة الاستثمار الأردن

Income Statement

Table 41 Income Statement

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Revenues										
Sales Revenues	93,750	191,250	195,075	198,977	202,956	207,015	211,155	215,379	219,686	224,080
Gross Operating Revenues	93,750	191,250	195,075	198,977	202,956	207,015	211,155	215,379	219,686	224,080
Operating Expenses	(86,732)	(109,472)	(112,115)	(114,824)	(117,601)	(120,448)	(123,367)	(126,360)	(129,428)	(132,574)
Gross Operating Profit	7,018	81,778	82,960	84,153	85,355	86,567	87,788	89,019	90,258	91,506
Gross Profit Percentage	*%7*	*%43*	*%43*	*%42*	*%42*	*%42*	*%42*	*%41*	*%41*	*%41*
Salaries and Benefits (Indirect	(21,600)	(22,248)	(22,915)	(23,603)	(24,311)	(25,040)	(25,792)	(26,565)	(27,362)	(28,183)
General and Administraive	(7,480)	(6,508)	(6,638)	(6,771)	(6,906)	(7,044)	(7,185)	(7,329)	(7,475)	(7,625)
Markeing Expenses	(2,018)	(2,058)	(2,100)	(2,142)	(2,184)	(2,228)	(2,273)	(2,318)	(2,364)	(2,412)
Pre Operating Expenses	(5,500)									
Gross Indirect Expenses	(36,598)	(30,814)	(31,653)	(32,515)	(33,401)	(34,312)	(35,249)	(36,212)	(37,202)	(38,219)
Income before Interest,	(29,580)	50,964	51,308	51,638	51,954	52,255	52,539	52,807	53,056	53,286
Fixed Assets Depreciations	(6,563)	(6,616)	(6,670)	(6,726)	(6,783)	(6,841)	(6,902)	(6,963)	(7,027)	(7,092)
Income before Tax and Interests	(36,144)	44,348	44,637	44,912	45,171	45,413	45,638	45,843	46,029	46,194
Bank Interests	(7,956)	(6,640)	(5,199)	(3,621)	(1,892)	0	0	0	0	0
Income Before Tax	(44,100)	37,708	39,439	41,291	43,279	45,413	45,638	45,843	46,029	46,194
Income Tax	0	0	0	0	0	0	0	0	0	0
Net Profit	(44,100)	37,708	39,439	41,291	43,279	45,413	45,638	45,843	46,029	46,194

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Net Profit Percentage	*%47-*	*%20*	*%20*	*%21*	*%21*	*%22*	*%22*	*%21*	*%21*	*%21*
Compulsory Reserves	0	(3,771)	(3,944)	(4,129)	(4,328)	(4,541)	(4,564)	(4,584)	(4,603)	(4,619)
Retained Earnings	(44,100)	(10,163)	25,332	62,494	101,445	142,316	183,390	224,649	266,075	307,650

Expected Cashflows Statement

The table below summarizes project cashflows statement, where net cash flows are positive during all years as below:

Table 42 Expected Cashflows

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Cash Inflows from Operating Activities										
Net Profit	(44,100)	37,708	39,439	41,291	43,279	45,413	45,638	45,843	46,029	46,194
Bank Interests	7,956	6,640	5,199	3,621	1,892	0	0	0	0	0
Depreciation	6,563	6,616	6,670	6,726	6,783	6,841	6,902	6,963	7,027	7,092
Total Operating Cashflows before	(29,580)	50,964	51,308	51,638	51,954	52,255	52,539	52,807	53,056	53,286
Inventory (Increase/Decrease)	(703)	(703)	(28)	(29)	(29)	(30)	(30)	(31)	(32)	(32)
Accounts Receviables (Increase/Decrease)	(15,625)	(16,250)	(638)	(650)	(663)	(677)	(690)	(704)	(718)	(732)
Accounts Paybales (Increase/Decrease)	0	0	0	0	0	0	0	0	0	0
Working Capital (Increase/Decrease)	(16,328)	(16,953)	(666)	(679)	(693)	(706)	(720)	(735)	(750)	(765)
Net Cashflows from Operating Activities	(45,908)	34,011	50,642	50,959	51,261	51,548	51,819	52,072	52,306	52,521
Cashflows from Investments Activities										

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Fixed Assets (Procurement)	(97,196)	(527)	(541)	(556)	(570)	(586)	(601)	(618)	(634)	(652)
Net Cashflows from Investments	(97,196)	(527)	(541)	(556)	(570)	(586)	(601)	(618)	(634)	(652)
Cashflows from Financing Activities										
Capital	83,751									
Loan Amortization	(13,856)	(15,172)	(16,613)	(18,191)	(19,920)	0	0	0	0	0
Bank Interest Rate	(7,956)	(6,640)	(5,199)	(3,621)	(1,892)	0	0	0	0	0
Loans	83,751	0	0	0	0	0	0	0	0	0
Net Cashflows from Financing Activities	145,691	(21,812)	(21,812)	(21,812)	(21,812)	0	0	0	0	0
Net (Increase/Decrease) in Cash	2,587	11,672	28,289	28,591	28,879	50,963	51,217	51,454	51,672	51,870
Cashflows at the Beginning of Period	0	2,587	14,258	42,547	71,139	100,018	150,980	202,198	253,652	305,324
Cashflows at the End of Period	2,587	14,258	42,547	71,139	100,018	150,980	202,198	253,652	305,324	357,194

Expected Balance Sheet

The balance sheet is one of the main statements which the project depends on, as it reflects the financial position of the entity and is usually estimated on the last day of the financial year.

All financial information and analysis of the project shall be the estimated for the years (2018-2027) as shown in the table below:

Table 43 Expected balance sheet

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Assets										
Current Assets										
Cash	2,587	14,258	42,547	71,139	100,018	150,980	202,198	253,652	305,324	357,194
Inventory	703	1,406	1,434	1,463	1,492	1,522	1,552	1,583	1,615	1,647
Accounts Receivable	15,625	31,875	32,513	33,163	33,826	34,503	35,193	35,896	36,614	37,347
Total current Assets	**18,914**	**47,539**	**76,494**	**105,764**	**135,336**	**187,004**	**238,942**	**291,131**	**343,553**	**396,187**
Non Current Assets										
Fixed Assets (net)	90,633	84,544	78,415	72,245	66,032	59,776	53,476	47,130	40,738	34,297
Total Non Current Assets	**90,633**	**84,544**	**78,415**	**72,245**	**66,032**	**59,776**	**53,476**	**47,130**	**40,738**	**34,297**
Total Assets	**109,547**	**132,083**	**154,908**	**178,009**	**201,368**	**246,781**	**292,418**	**338,261**	**384,291**	**430,485**
Liabilities	-	-	-	-	-					
Current Liabilities										
Paybles	0	0	0	0	0	0	0	0	0	0
Remaining amount of Loan	15,172	16,613	18,191	19,920	0	0	0	0	0	0
Total current Liabilities	**15,172**	**16,613**	**18,191**	**19,920**	**0**	**0**	**0**	**0**	**0**	**0**
Non Current Liabilities										
Long Terms Loans	54,724	38,111	19,920	0	0	0	0	0	0	0
Total Long Term Liabilities	**54,724**	**38,111**	**19,920**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Total Liabilities	**69,896**	**54,724**	**38,111**	**19,920**	**0**	**0**	**0**	**0**	**0**	**0**
Owners Equity										

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Shareholders Contributions	83,751	83,751	83,751	83,751	83,751	83,751	83,751	83,751	83,751	83,751
Statutory Reseve	0	3,771	7,715	11,844	16,172	20,713	25,277	29,861	34,464	39,083
Retained Profits	(44,100)	(10,163)	25,332	62,494	101,445	142,316	183,390	224,649	266,075	307,650
Total Equity	39,651	77,359	116,798	158,089	201,368	246,781	292,418	338,261	384,291	430,485
Total Liabilities and Equity	**109,547**	**132,083**	**154,908**	**178,009**	**201,368**	**246,781**	**292,418**	**338,261**	**384,291**	**430,485**

Feasibility Indicators

There are more than one method to calculate the discounted cash flow statement. However, in this study, WACC was used to calculate the discounted cash flows and net investment value. The Company's financing structure is based on equity and loans; therefore, the discounted rate should be adjusted according to WACC based on the following assumptions, the following table illustrates the net expected free cash flow of the project:

Table 44 Free Net Cashflows

Net Cashflows	Before Operating	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027	Final value
Net Free Cashflows	(167,503)	40,408)	33,484	50,101	50,403	50,691	50,963	51,217	51,454	51,672	51,870	677,271
Discount Factor	1.00	0.89	0.80	0.71	0.64	0.57	0.51	0.45	0.40	0.36	0.32	0.32
Net Present Value for Cash Flows	**(167,503)**	**36,090)**	**26,710**	**35,694**	**32,072**	**28,808**	**25,868**	**23,219**	**20,834**	**18,686**	**16,753**	**218,746**

The table below illustrates payback period for the project, as it is one of the indicators which investors care about before taking the investment decision as they need to know when the project will return the invested amount of money. Payback period definition is the total required period so that the project generates a total money equal to the total invested capital expenditure, as investor is always looking to return the full invested amount of money at the earliest possible.

Table 45 Payback Period

Payback Period	Before Operating	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Net Free Cashflows	(167,503)	(40,408)	33,484	50,101	50,403	50,691	50,963	51,217	51,454	51,672	51,870
Project Value Returned	167,503	207,911	174,427	124,326	73,923	23,232	(27,730)	(78,948)	(130,402)	(182,074)	(233,944)
Payback Period (Year)	5	1	1	1	1	1	0	0	0	0	0

Table 46 Financial Analysis Results

Weighted Average Cost of Capital (WACC)	%11.97
Net Present Value for Cashflows	243,797
Payback Period	5
Internal Rate of Return	%23.83

Sensitivity Analysis

Sensitivity analysis is conducted to investigate the effects of changes in significant inputs of the financial analysis. This includes changes in revenues, total investment costs, and operating costs.

Table 47 Sensitivity Analysis

Sensitivity Analysis	Internal Rate of Return	Payback Period	WACC	Sensitivity Analysis
Original Scenario	23.83%	5	244	11.97%
Revenues declined by 10%	14.38%	8	51	11.97%
Operating Expenses Increased by 10%	17.31%	7	109	11.97%
Total cost Increased by 10%	16.09%	7	92	11.97%
Total cost Increased and revenues decreased by 10%	4.13%	10	-101	11.97%

Conclusions and Recommendations

Based on the profitability analysis and considering that project internal rate of return is higher than WACC, it is concluded that the project is feasible as the net present value for the project is positive 243,797 Jordan Dinars considering that the project provides 15 Job Opportunities for the governorate residents.

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